

MAT 2440 – Discrete Structures

Review Problems for Exam 1

Fall 2026

Exam information. Exam 1 is designed to be completed in approximately 100 minutes. Calculators and electronic devices may be used during the exam.

The following problems are intended to help you review the main ideas and types of problems that may appear on the exam. They are not intended to be an exact copy of the exam.

Review Problems

1. System specifications

Consider a computer system with the following specifications:

- “If the server is running, then the database is available.”
- “If the database is available, then users can log in.”
- “The server is running.”
- “Users cannot log in.”

Let

p : “The server is running,”

q : “The database is available,”

and

r : “Users can log in.”

- (a) Express each specification using p , q , r , and logical connectives.
- (b) Determine whether the system specifications are *consistent*; that is, determine whether there is an assignment of truth values to p , q , and r for which all four specifications are true.
- (c) Justify your conclusion using a truth table or logical reasoning.

2. Translating statements into propositional logic

Let

p : “The student studies for the exam,”

q : “The student attends the review session,”

and

r : “The student passes the exam.”

Write each statement using p , q , r , and logical connectives.

- (a) The student studies for the exam and attends the review session.

- (b) If the student studies for the exam, then the student passes the exam.
- (c) The student passes the exam only if the student studies for the exam.
- (d) If the student studies for the exam or attends the review session, then the student passes the exam.
- (e) The student does not study for the exam, but passes the exam.

3. Evaluating compound propositions

Suppose that p is true, q is false, and r is true. Determine the truth value of each proposition.

- (a) $p \wedge q$
- (b) $p \vee q$
- (c) $q \rightarrow r$
- (d) $(p \vee q) \rightarrow r$
- (e) $\neg p \vee (q \wedge r)$

Show enough work to justify your answers.

4. Truth tables

Construct a complete truth table for

$$(p \vee q) \rightarrow \neg r.$$

Your table should include columns for the intermediate expressions needed to evaluate the compound proposition.

Based on your truth table, determine whether the proposition is a

tautology, contradiction, or contingency.

5. Logical equivalence using truth tables

Use a truth table to determine whether

$$(p \rightarrow r) \wedge (q \rightarrow r)$$

and

$$(p \vee q) \rightarrow r$$

are logically equivalent.

Clearly explain how your truth table supports your conclusion.

6. Conditional statements

Consider the statement:

“If an integer is divisible by 6, then it is divisible by 3.”

- (a) Write the converse.
- (b) Write the inverse.
- (c) Write the contrapositive.
- (d) Which of these statements is logically equivalent to the original conditional?

7. Logical equivalences

Show that

$$\neg(p \vee (\neg p \wedge q))$$

is logically equivalent to

$$\neg p \wedge \neg q.$$

Use a sequence of logical equivalences rather than a truth table.

For each step, state the law that you used. You may use laws such as De Morgan’s laws, the double negation law, distributive laws, commutative laws, associative laws, identity laws, domination laws, and negation laws.

8. Tautology, contradiction, or contingency

For each proposition below, determine whether it is a tautology, a contradiction, or a contingency. Justify your answer using either a truth table or logical equivalences.

- (a) $p \vee \neg p$
- (b) $p \wedge \neg p$
- (c) $(p \wedge q) \rightarrow p$
- (d) $p \rightarrow q$

9. Predicates and quantifiers

Let the domain be

$$U = \{-3, -2, -1, 0, 1, 2, 3\},$$

and let

$$P(x) : x^2 < 5.$$

Determine the truth value of each statement. Justify your answer. For an existential statement that is true, give a value of x that makes it true. For a universal statement that is false, give a counterexample.

- (a) $P(2)$

- (b) $P(3)$
- (c) $\exists x P(x)$
- (d) $\forall x P(x)$
- (e) $\exists x \neg P(x)$

10. Translating and negating quantified statements

Let the domain consist of all students at a college.

Define

$$M(x) : \text{"}x \text{ is a mathematics major,"}$$

and

$$C(x) : \text{"}x \text{ has taken a computer science course."}$$

- (a) Translate the following statement into predicate logic:

“Every mathematics major has taken a computer science course.”

- (b) Translate the following statement into predicate logic:

“Some mathematics major has taken a computer science course.”

- (c) Negate the statement

$$\forall x (M(x) \rightarrow C(x)).$$

Move the negation as far inward as possible.

- (d) Write your answer to part (c) in ordinary English.

Use these problems to identify topics that require additional practice before Exam 1.